



Welcome to CSC 365

Internet Programming

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What is CSC 365 about?

Web Development

...but what is Web Development?

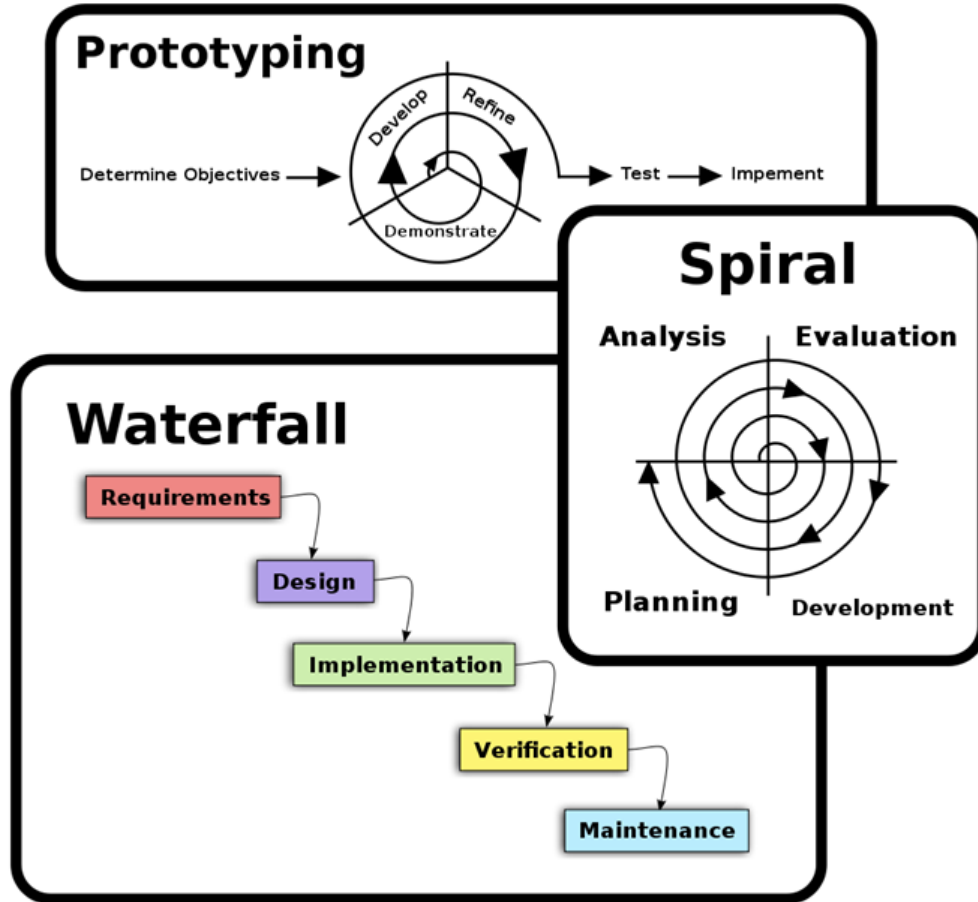
Web Development is the process of creating and maintaining websites and web applications that run on the internet. It combines programming, design, and infrastructure to deliver user-facing experiences and backend functionality.

What does a Software Engineer do?

Definition: A software engineer is a person who applies the principles of software engineering to the design, development, maintenance, testing, and evaluation of computer software.

But how does a Software Engineer go about this?

Software Development Lifecycle

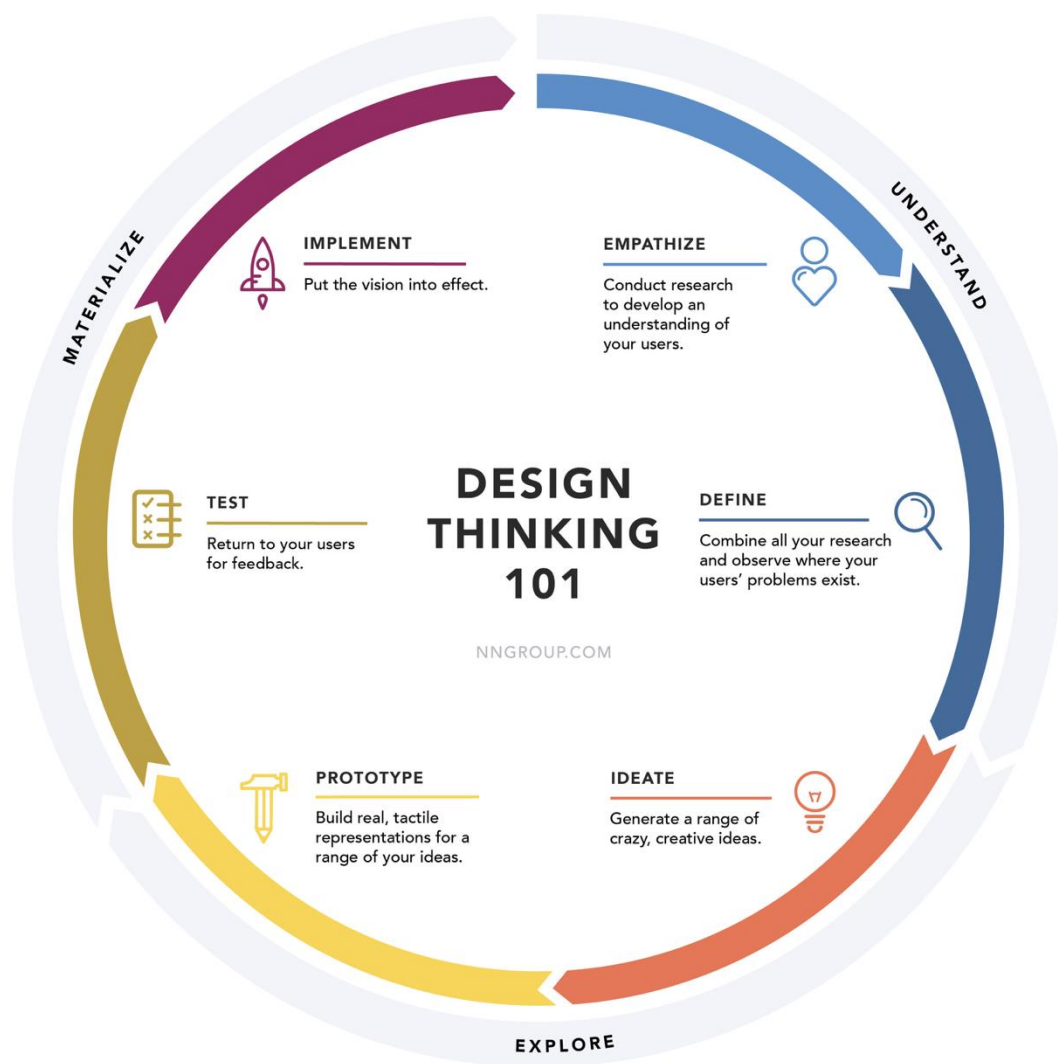


What does a UX Designer do?

Definition: User experience (UX) design is the process that design teams use to create products that provide meaningful and relevant experiences to users. A UX designer is concerned with the entire process of acquiring and integrating a product, including aspects of branding, design, usability, and function.

How does a UX designer go about this?

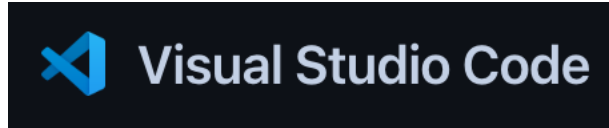
UX Design Process



UX Development Trajectories

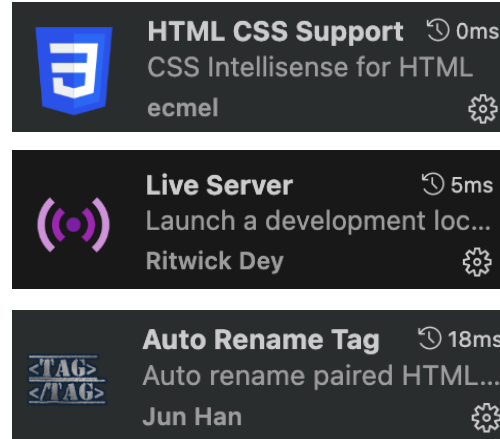
- One-person development team to build full-stack applications
- A developer who speaks the language of the designers
- A designer who can also build native prototypes
- A bridge/translator between designers and developers in large/complex organizations

VS code



<https://code.visualstudio.com/>

Extensions:



The Web.

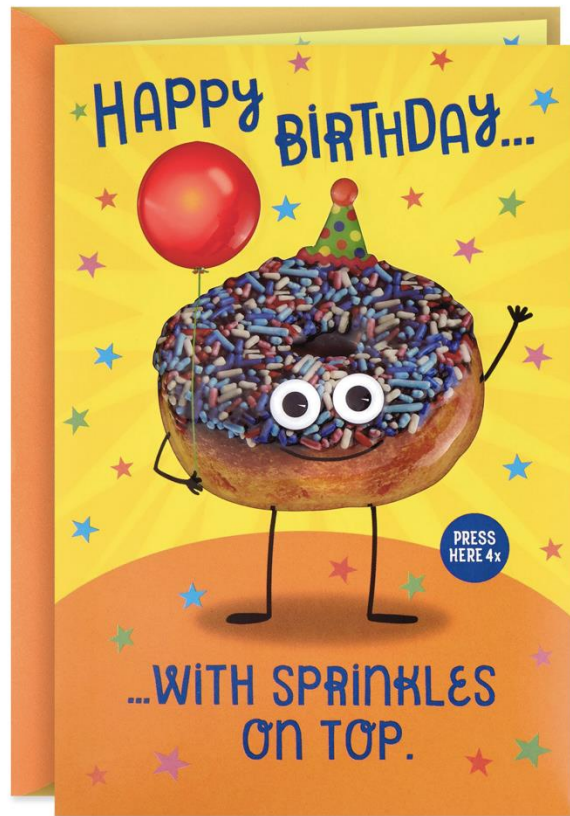
- A web application or web app is a client–server computer program that the client (including the user interface and client-side logic) runs in a web browser. (wiki)
- Two keywords: client-server and browser.

The Web.

A Website is like a birthday card!

Concept	Behavior
HTML	Structure
CSS	Design
JS	Behavior

How does it get delivered? HTTP!

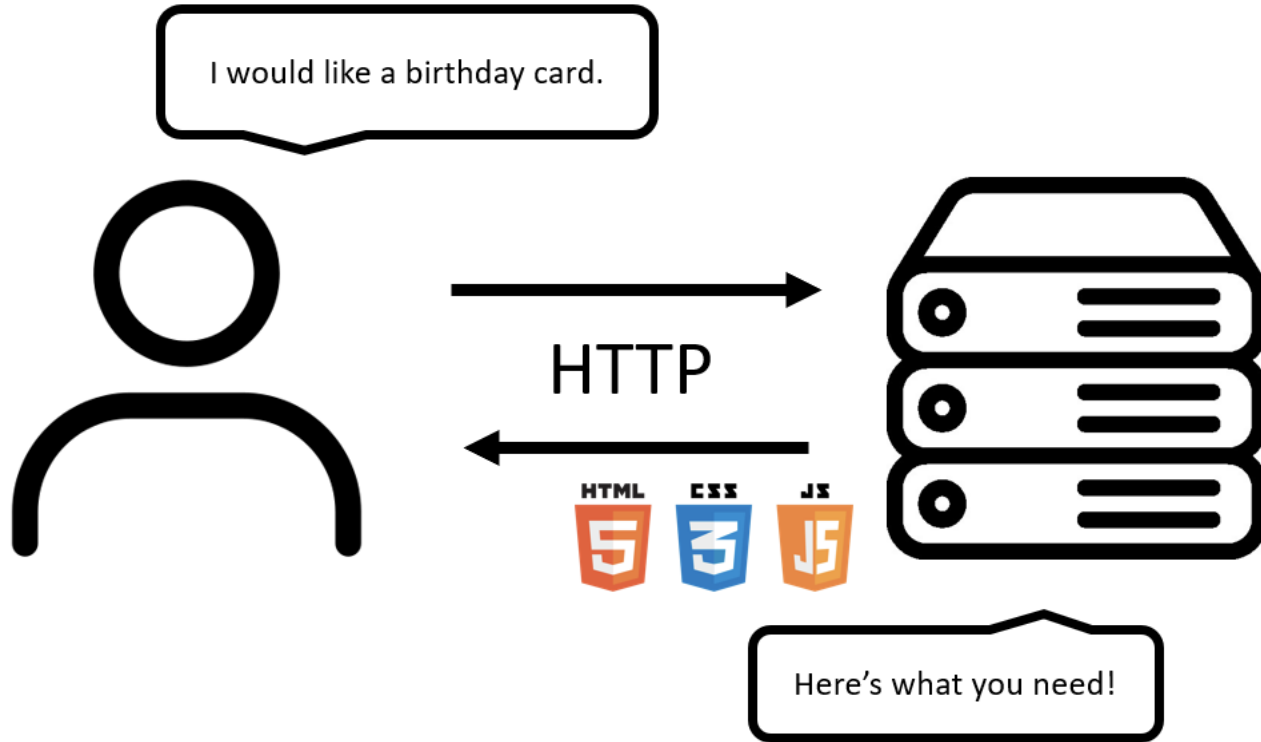


Browser

- A web browser (commonly referred to as a browser) is a software application for accessing information on the World Wide Web. (wiki)
- Later when we reach JavaScript, I will also show you that a browser can be a compiler and debugger.



HTTP: Hypertext Transfer Protocol

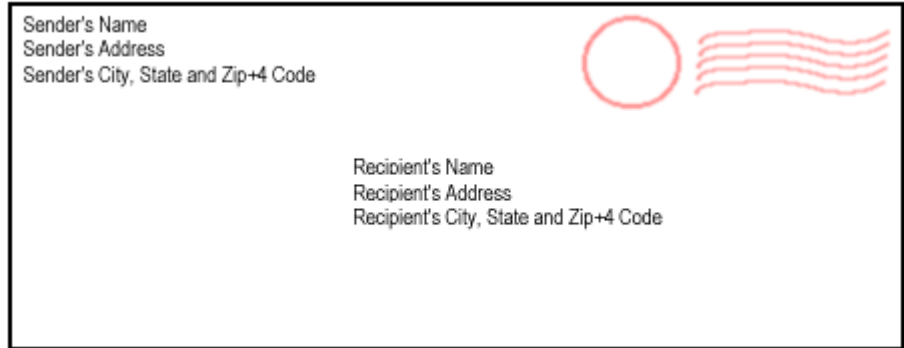


World Wide Web (WWW)

- The World Wide Web Consortium is the main international standards organization for the World Wide Web.
- We will use it for some definitions.
- <https://www.w3.org/>

Uniform Resource Locators (URLs)

- Uniform Resource Identifiers (URIs, aka URLs) are short strings that identify resources in the web: documents, images, downloadable files, services, electronic mailboxes, and other resources.
- It is similar to a mailing address.



Uniform Resource Locators (URLs)

- More details, see <http://www.w3.org/Addressing/>
- Examples:
 - <https://www.amazon.com/>
 - <https://canvas.ecu.edu/>
 - <https://computerscience.missouristate.edu>

Domain Name

- A domain name is an identification string that defines a realm of administrative autonomy, authority or control within the Internet (Wiki).
- For example,
 - URL: www.example.com/index
 - Domain name: example.com

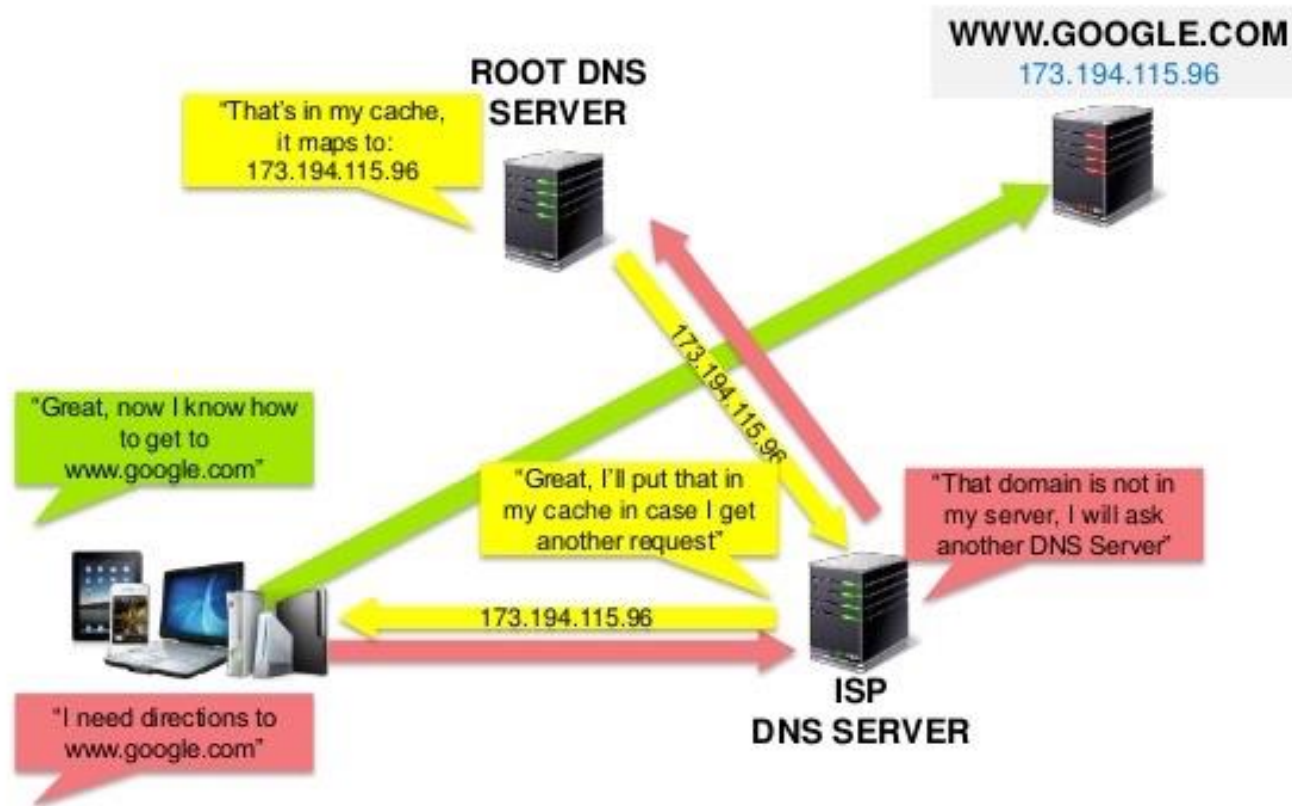
IP Address

- An Internet Protocol address (IP address) is a numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication (Wiki).
- A unique identifier given to every machine that connects to the internet.
- Typical IPv4 addresses resemble the following: 192.x.x.x while IPv6 addresses look similar to: fdcd:xxxx:xxxx::/48.
- It is also similar to a mailing address.

Domain Name System

- Servers do not recognize your domain name.
- Banner ID (IP address) vs Name (domain name)
- The Domain Name System (DNS) is a hierarchical decentralized naming system for computers, services, or other resources connected to the Internet or a private network (Wiki).
- It can provide you IP address based on a domain name (similar to your contact app).

Domain Name System



Why not just IP? Then we do not need DNS.



Domain Name System

Ans: Because it is hard to

remember... Also, not enough IP addresses: It is often case that one web server will handle sites on the same IPv4 address (different ports). However, sometimes, you need to use IP instead of URL:

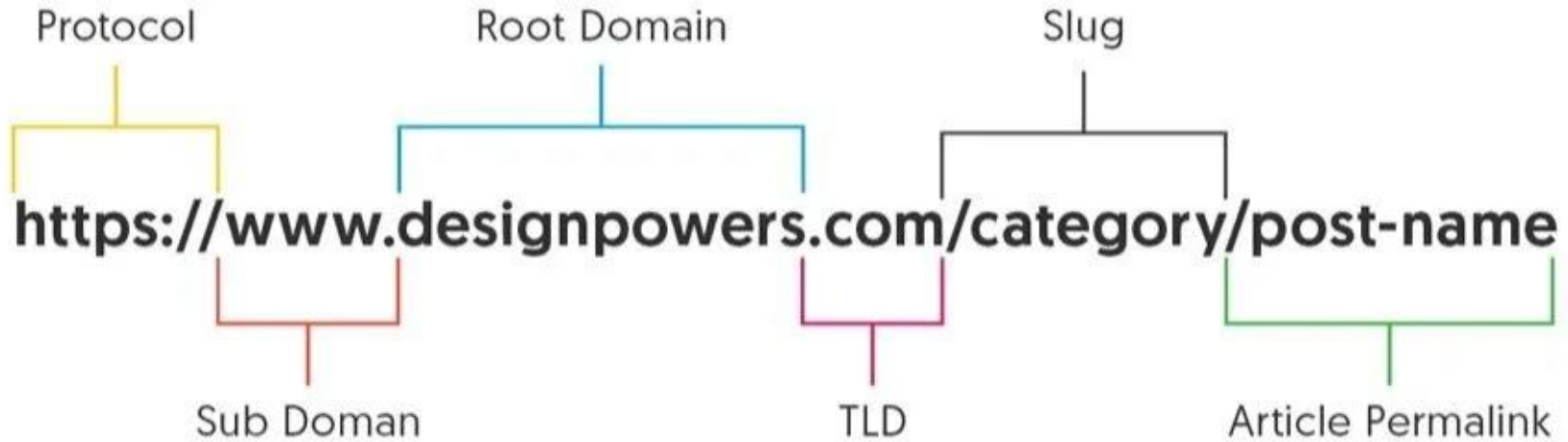
- If a website's DNS system is down then you may still be able to access it via its IP address. This occurred during Twitter's large outage back in 2016.
- A particular system may only accept IP addresses, therefore you need to convert your URL to IP.
- You want to find out more information about the URL including who owns it, where it resides, or simply check if the domain record is correct.

Find your IP address:

Windows: cmd -> ipconfig

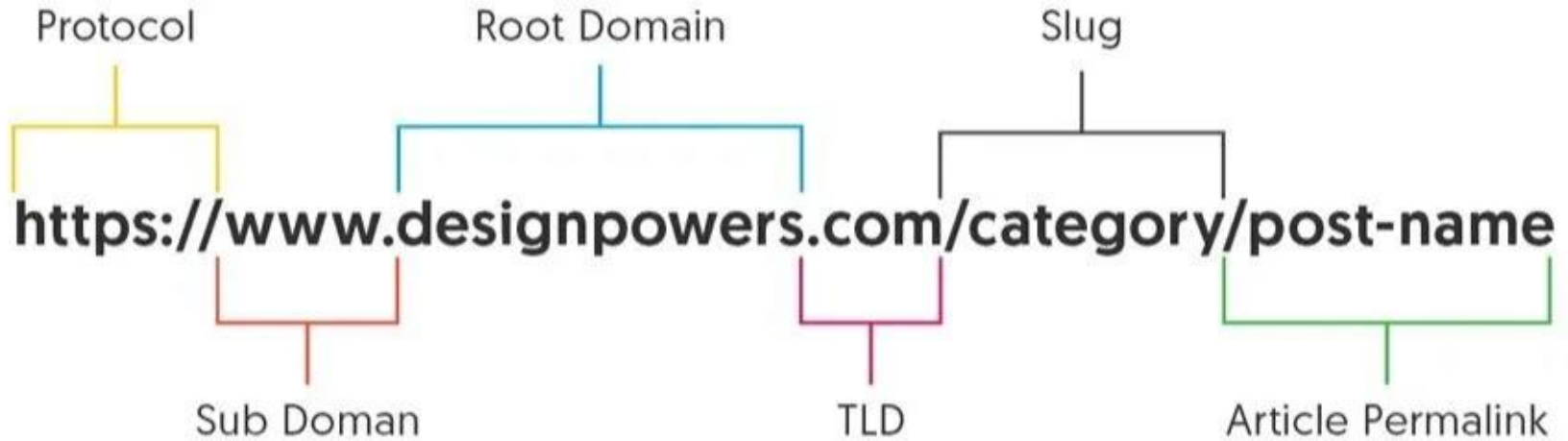
Ubuntu: ip addr

How are URLs structured?



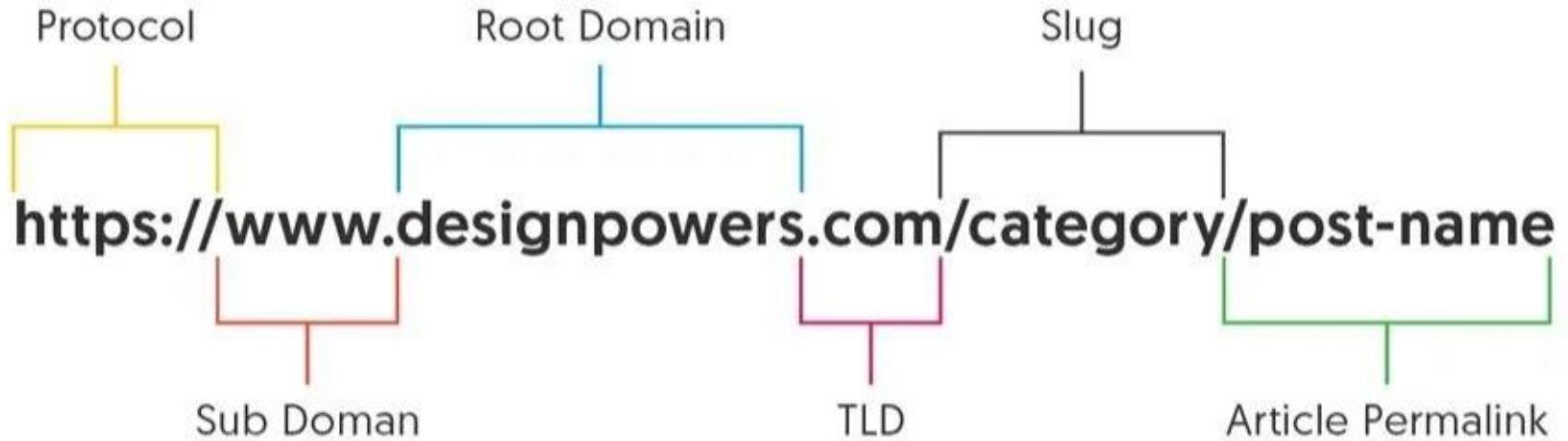
Protocol: Specifies the method used to access the resource (e.g., `http://` or `https://`).

How are URLs structured?



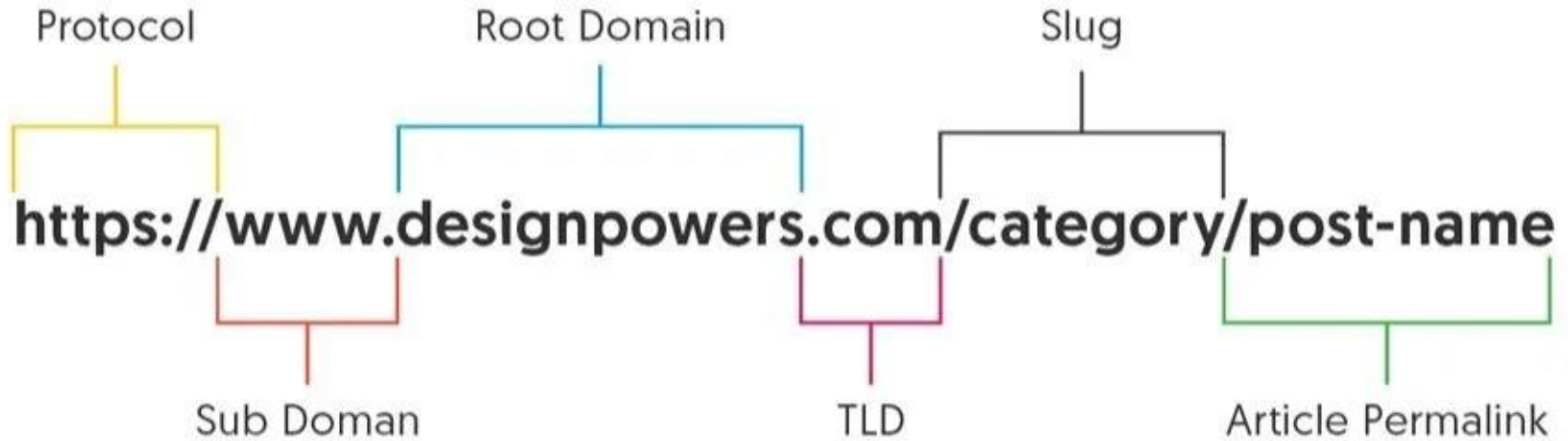
// indicates this is a “hierarchical URL” required for hierarchical schemes, like http not required for schemes like mailto

How are URLs structured?



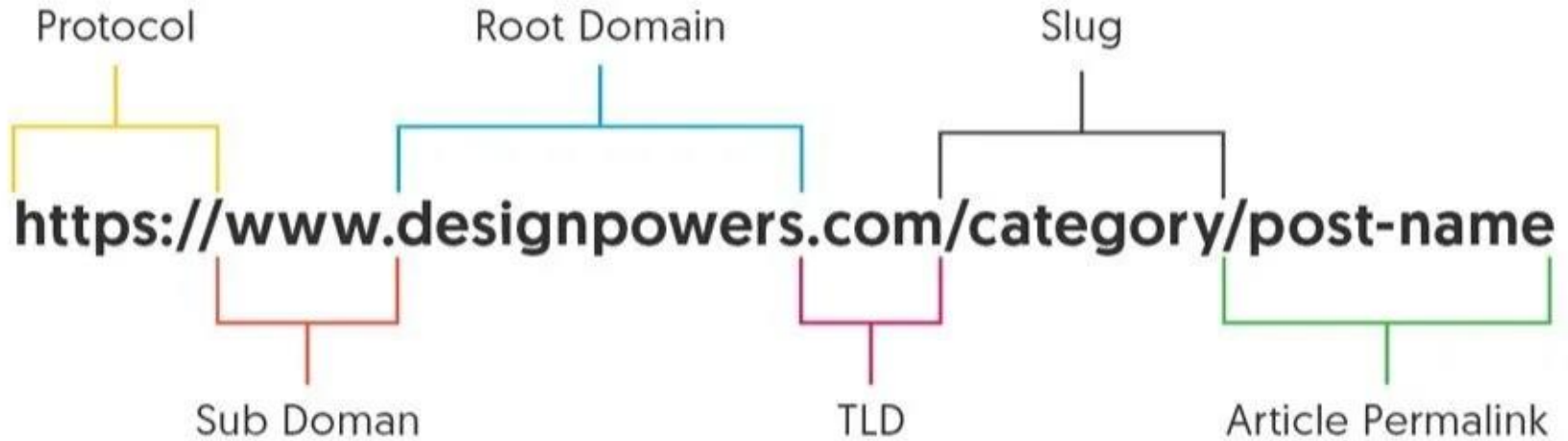
Subdomain: An optional part that precedes the domain name (e.g., `www`).

How are URLs structured?



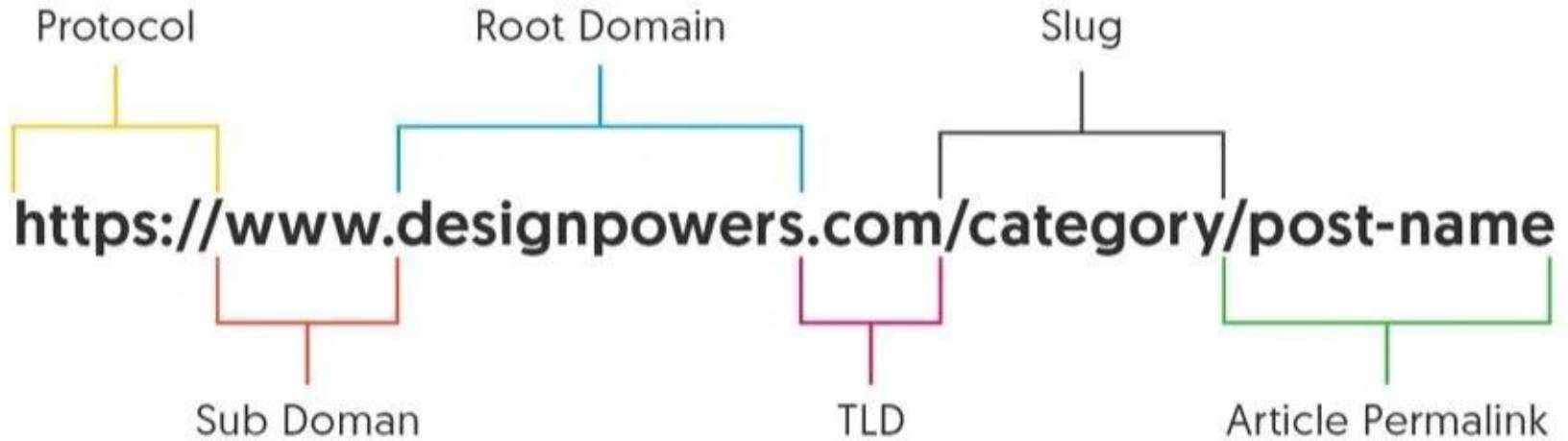
Domain Name: The unique address of a website.

How are URLs structured?



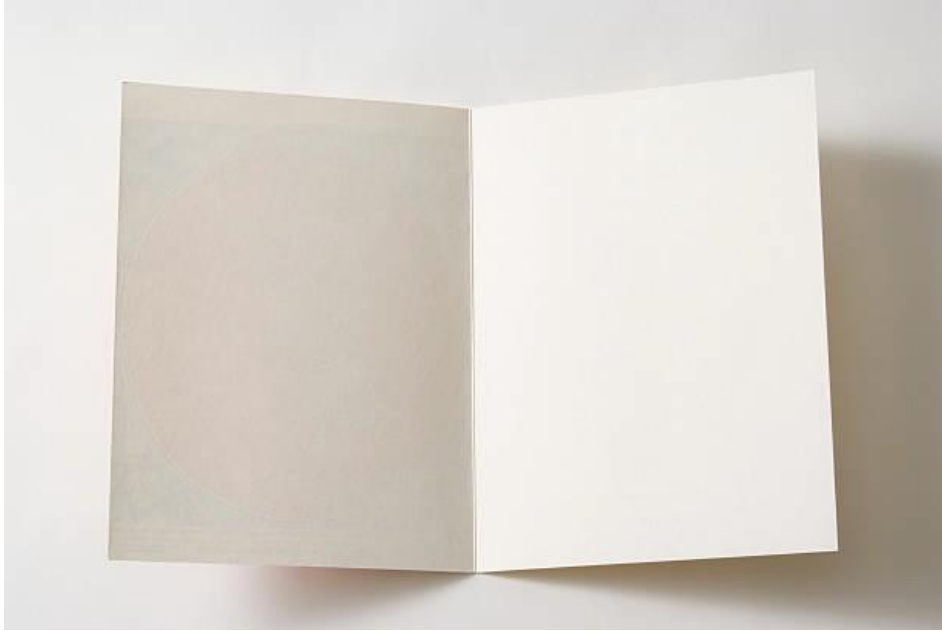
Top-Level Domain (TLD): The last part of the domain name (e.g., `.com`, `.org`, `.edu`).

How are URLs structured?



Path/to/resource: Specifies the location of a specific page or file within the website (e.g., `/about/contact`). .

Start: nothing inside of the card

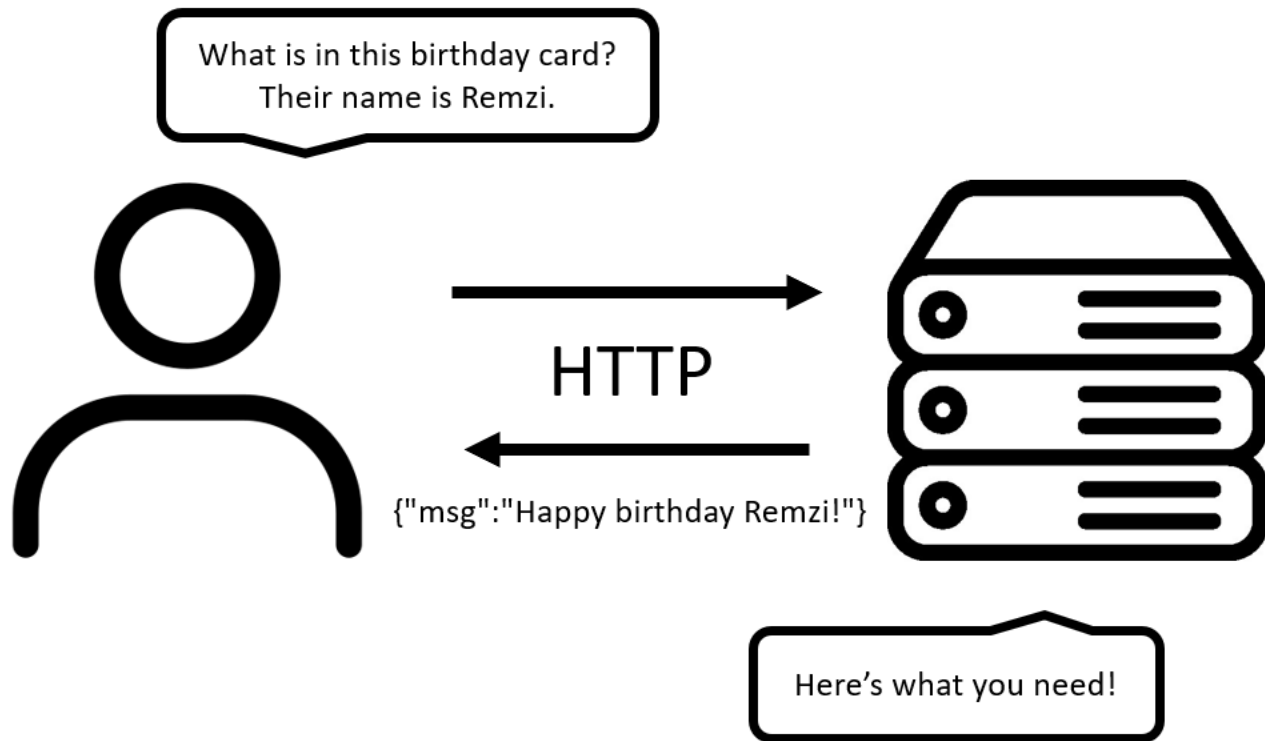


Communication: JSON

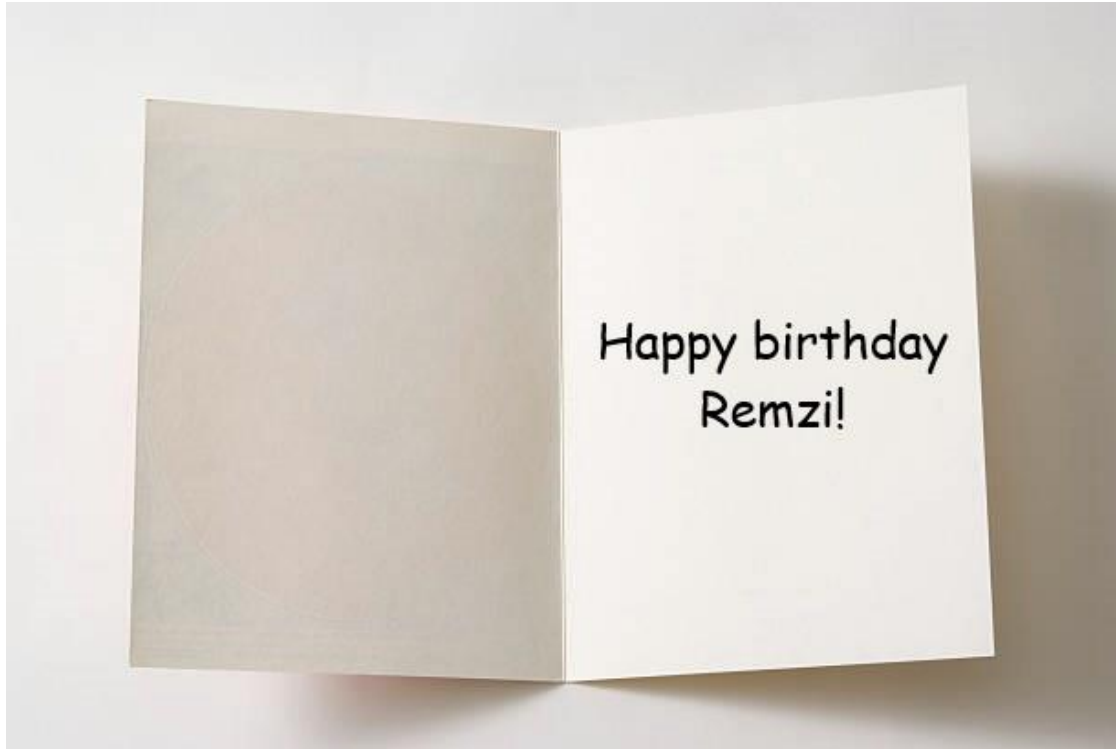
We need a structured way to communicate data.
We'll use JSON.

```
[
  {
    "id": 1,
    "title": "New interactive mural invites
              exploration and engagement with
              science",
    "img": "https://www.coletnelson.us
            /cs571/f22/hw7/api/news/images
            /exhibit",
    "tags": [
      "science",
      "entertainment"
    ]
  },
  {
    "id": 2,
    "title": "Sewer sleuthing, air
```

HTTP



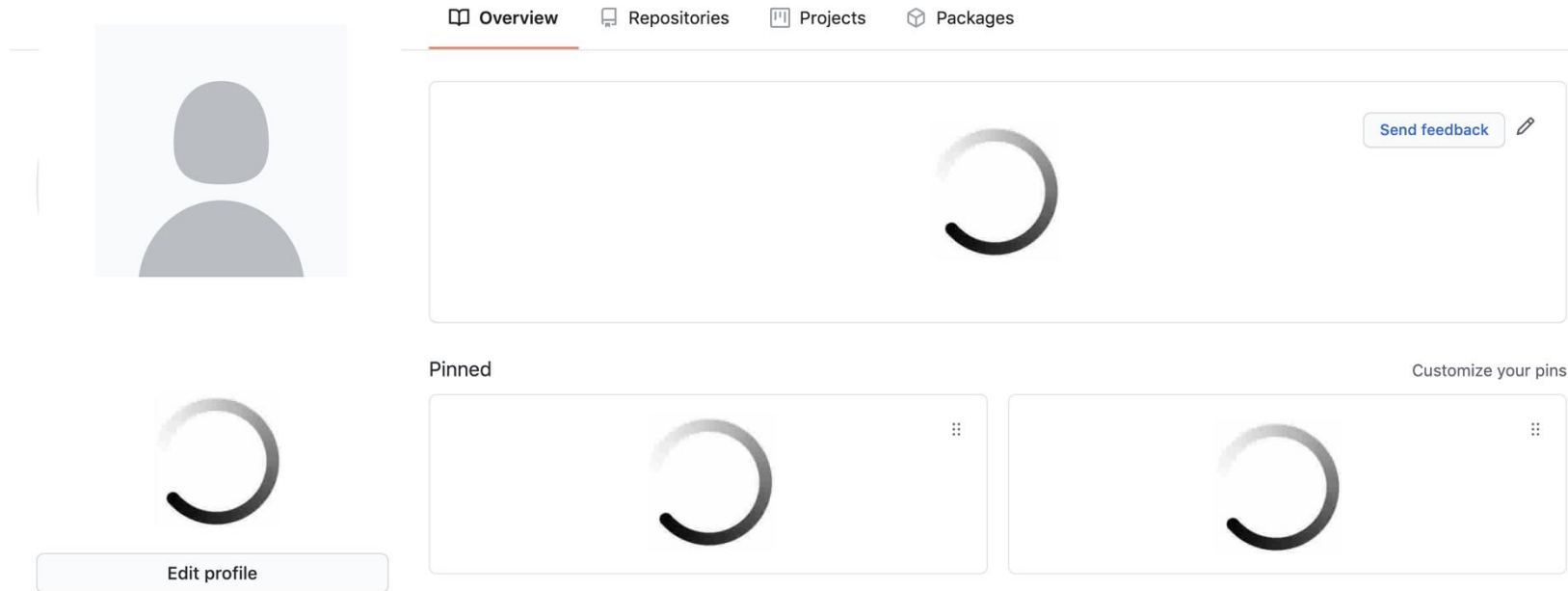
Start: message on the card



Better!

Real-World Example

How does GitHub display profile webpages?



What can be displayed with HTML/CSS/JS
on load...

Real-World Example

```
HTTP/1.1 200 OK
Server: GitHub.com
Date: Fri, 20 Jan 2023 21:36:04 GMT
Content-Type: application/json; charset=utf-8
Cache-Control: public, max-age=60, s-maxage=60
Vary: Accept, Accept-Encoding, Accept, X-Requested-With
Last-Modified: Fri, 20 Jan 2023 12:09:09 GMT
```

```
{
  "login": "ctnelson1997",
  "id": 37457808,
  "node_id": "MDQ6VXNlcjM3NDU3ODAA4",
  "avatar_url": "https://avatars.githubusercontent.com/u/37457808?v=4",
  "gravatar_id": "",
  "url": "https://api.github.com/users/ctnelson1997",
  "public_repos": 6,
  "public_gists": 1,
  "followers": 3,
  "following": 3,
  "created_at": "2018-03-16T20:56:14Z",
  "updated_at": "2023-01-20T12:09:09Z"
}
```

Response Code

Response
Headers

Response
Body

JSON data received!

Real-World Example



Mona Lisa Octocat
octocat

Hi, I'm Mona 🙋 You might recognize me as [@github](#)'s mascot 🐱🐱

Edit profile

Overview Repositories Projects Packages

octocat / README.md [Send feedback](#) ✎

Hi there 🙋

- 🚧 I'm currently working on something cool!
- 🌱 I'm currently learning with help from [docs.github.com](#)
- 💬 Ask me about GitHub

Pinned Customize your pins

 **atom** ⋮

Forked from atom/atom

 The hackable text editor

JavaScript

 **vscode** ⋮

Forked from microsoft/vscode

Visual Studio Code

TypeScript

Why JSON?

- easy to understand
- human-readable
- language agnostic
- easily convertible to JS objects

[Try json](#)

```
[
  {
    "id": 1,
    "title": "New interactive mural invites
              exploration and engagement with
              science",
    "img": "https://www.coletnelson.us
            /cs571/f22/hw7/api/news/images
            /exhibit",
    "tags": [
      "science",
      "entertainment"
    ]
  },
  {
    "id": 2,
    "title": "Sewer sleuthing, air
```

JSON Basics

- Valid value types include...
 - string e.g. "hello world"
 - number e.g. 24
 - boolean e.g. false
 - array e.g. [1, true, 2, 4.2, "smith"]
 - object e.g. {"name": "Carl", "age": 24}
 - null
- Values can be nested!

In Class Activity 1

Expressing yourself in JSON.

Next Class

Week 1 Introduction & Getting Started:

